

An Evaluation Metric for Single Image-to-3D Models Based on Object Detection Perspective

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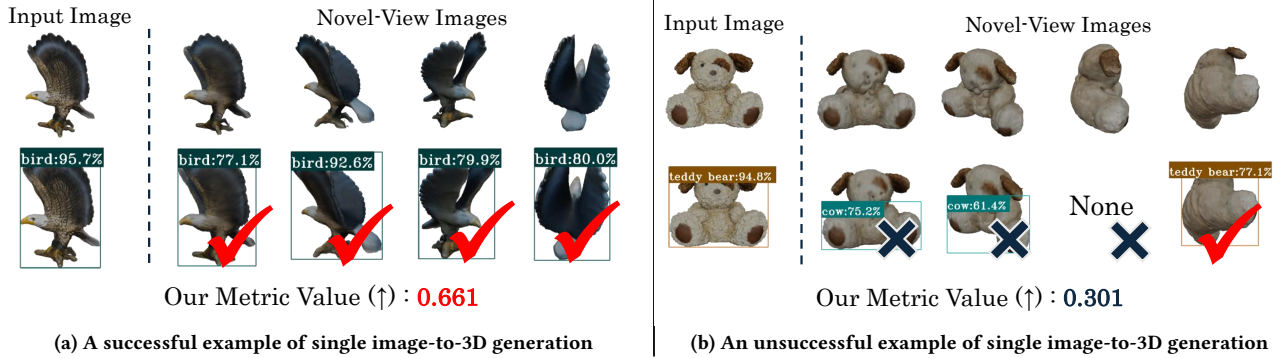


Figure 1: Our proposed metric enables evaluations of "single image-to-3D" models considering semantic information and focusing especially on objects with only one input image. The proposed metric performs evaluations by comparing object detection results of the input image and the novel-view images. The novel-view images of the left example (a) are recognized as the same-class object as the input image from any angle, resulting in a high metric value. On the other hand, those of the right example (b) are sometimes recognized as a different-class object or not recognized as an object, resulting in a low metric value.

Abstract

In this paper, we propose a novel evaluation metric for "single image-to-3D" models based on object detection perspective. Unlike conventional metrics that require the original 3D model or a lot of multi-view images of the model as reference, the proposed metric allows evaluations of the "single image-to-3D" models with just one input image. The proposed evaluation metric can be calculated by comparing the object detection results of the input image and novel-view images rendered from the generated object. Therefore, this metric enables evaluations that consider semantic information and focus especially on objects in the rendered images. The proposed evaluation metric is validated by the rank correlation coefficient between the order of its values and the order of the "single image-to-3D" models' publication dates. The experimental results show

that our proposed metric enables semantically accurate evaluations using only one input image. The code of this work is available at <https://github.com/EvalSingleImg23D/EvalSingleImg23D>.

CCS Concepts

• General and reference → Metrics; Evaluation; • Computing methodologies → Reconstruction; Object detection.

Keywords

Evaluation, Single Image-to-3D, Generation, Object Detection

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1 Introduction

Generating 3D objects from a single image, known as "single image-to-3D", is a challenging task that requires a machine to imagine 3D shapes from a 2D image. Despite the difficulty, single image-to-3D technology has achieved remarkable progress and has attracted



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more attention due to its potential for wide range of applications, including games, animations, extended reality, art and robotics. In the course of single image-to-3D technology development, evaluation metrics always play an important role. They help us quantitatively evaluate how far a new single image-to-3D model is superior to conventional ones and determine which model to adopt for the above applications.

Conventional metrics are classified into the following three categories: pixel value-based, structure-based, and deep learning model-based. To calculate these metrics, some assets are required, such as the input image, the original 3D model (i.e., the 3D model of the object in the input image), reference images (i.e., multi-view images of the original 3D model), the generated 3D model, and novel-view images (i.e., multi-view images of the generated object). The pixel-based metrics include Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) [Wang et al. 2004]. They compare the luminance values of each pixel between the novel-view images and the corresponding reference images. The structure-based metrics include Chamfer Distance (CD) [Borgefors 1986], Volume Intersection over Union (IoU) and F-score, which compare the coordinates of point clouds or the shape of voxels between the generated 3D model and the original 3D model. The deep learning model-based metrics include Learned Perceptual Image Patch Similarity (LPIPS) [Zhang et al. 2018] and Contrastive Language-Image Pre-training similarity (CLIP-similarity) [Radford et al. 2021], which compare the embedded representations from deep learning models between the novel-view images and the corresponding reference images. Although the pixel-based and the structure-based metrics are widely used for their simplicity, they lack semantic considerations of the objects. In contrast, the deep learning-based metrics can incorporate semantic information and recently they have even become the mainstream.

Deep learning-based metrics allow for evaluations that align with human perception by considering semantic information in deep neural networks. However, these metrics treat the entire image equally, rather than focusing on specific objects in the image. In practical applications, the quality of the main object may often be considered to be more important than that of the entire composition, such as the background. Therefore, there is room to establish another evaluation metric especially designed to focus on objects. In addition, CLIP-similarity lacks a theoretical basis for image-to-image similarity evaluation since CLIP was originally designed to optimize correspondence between text and images rather than between images. Therefore, the new metric would need to employ models that provide both visually and semantically accurate latent representations for objects in images. Furthermore, most of the conventional metrics require the original 3D model or the reference images for evaluation. However, the requirement for these additional materials, which are unnecessary for the generation process, is not desirable for realistic scenarios in practical use.

In this paper, we propose a novel evaluation metric for single image-to-3D models from object detection perspective. Our metric aims to evaluate the quality of the generated objects using confidence scores obtained by object detection models. As shown in Fig. 1, by comparing detection results of the input image and those of the novel-view images, the proposed metric checks whether the generated object is identified as the same-class as the original object

in the input image. Since object detection models focus only on the contents within the detected objects' bounding boxes and not on the entire image, it is possible to evaluate by concentrating solely on the objects. Also, this metric takes into account semantic similarity among the detected class names using ELECTRA [Clark et al. 2020] word embeddings to accurately and quantitatively evaluate how the generated object is semantically distant from the original object in the input image. Furthermore, unlike conventional metrics that require the original 3D model or reference images, the proposed metric can perform evaluations using only one image: the input image to single image-to-3D models itself. In this way, the proposed metric enables semantically accurate evaluations especially focusing on objects while using only one input image.

In summary, our contributions are shown as follows:

- We propose a novel evaluation metric for single image-to-3D models with the object detection that especially focuses on objects and considers semantic contents.
- The proposed metric enables evaluations from only a single input image, which aims to be suitable for realistic scenarios.

2 Proposed Evaluation Metric

The proposed metric consists of three-step processes: (1) generation and rendering, (2) estimation of confidence scores and (3) calculation of evaluation metric.

2.1 Single Image-to-3D Generation and Rendering from Diverse Viewpoints

We input an image I to a single image-to-3D model to be evaluated and generate a 3D object. Then we render novel-view images of the generated object at the same degree τ intervals for elevation and azimuth, and denote each novel-view image as i . Although conventional evaluations of single image-to-3D models used to be performed only on the equatorial plane without considering elevation angles, our proposed metric considers them in order to accurately evaluate the quality of generated 3D objects.

2.2 Estimation of Confidence Scores via Object Detection Model

We use an object detection model $\mathcal{M}$ to both the input image I and the novel-view image i , obtaining confidence scores $\mathbf{p}_I$ and $\mathbf{p}_i$. From each detection result, we extract the classes with confidence scores in the top five and denote this set as C_I and C_i , and the union of C_I and C_i as $C_{I \cup i}$. We use only top five classes since the confidence scores of those ranked 6th and below are usually very low (around 1.0×10^{-5}) and are not essential for the evaluation. Next, we extract the confidence scores of the classes in $C_{I \cup i}$ from $\mathbf{p}_I$ and $\mathbf{p}_i$. These scores are then normalized using the softmax function, resulting in $\mathbf{p}'_I$ and $\mathbf{p}'_i$, respectively. The temperature parameter of the softmax function is configured to avoid excessive smoothing of the confidence scores. By considering not only the most confident class of the object detection results but also those of the top five classes, it becomes possible to accurately represent the features of the object to be evaluated. Furthermore, this approach allows to mitigate large changes in the evaluation metric values that may result from changes in the detected classes.

2.3 Calculation of Evaluation Metric

First, we calculate local index values at each viewpoint. For each novel-view image i , its local index value r_i is defined as the cosine similarity between $\mathbf{p}'_I$ and $\mathbf{p}'_i$, weighted by semantic similarity among the detected classes. The local index value r_i for novel-view image i is given by the following equation:

$$r_i = \frac{\mathbf{p}'_I^\top \mathbf{A}_i \mathbf{p}'_i}{\|\mathbf{p}'_I\| \|\mathbf{p}'_i\|}, \quad (1)$$

where $\mathbf{A}_i$ is a diagonal matrix representing semantic similarity. The diagonal elements of $\mathbf{A}_i$ are defined as the maximum value of the cosine similarity between the embedding vectors of the class names belonging to $C_{I \cup i}$ and those belonging to C_I . For each element $\{c_1, c_2, \dots, c_j, \dots\} \in C_{I \cup i}$ and $\{c'_1, c'_2, \dots, c'_k, \dots\} \in C_I$, we obtain a word embedding vector $\mathbf{e}_j$ and $\mathbf{e}_k$ from ELECTRA [Clark et al. 2020]. Consequently, the diagonal elements of $\mathbf{A}_i$ is represented as follows:

$$A_i(j, j) = \max_{1 \leq k \leq |C_I|} \frac{\mathbf{e}_j^\top \mathbf{e}_k}{\|\mathbf{e}_j\| \|\mathbf{e}_k\|} \quad (j = 1, 2, \dots, |C_{I \cup i}|). \quad (2)$$

Note that r_i is always 1 or less since the elements of $\mathbf{A}_i$ do not exceed 1. Also, when no object is detected in i , r_i is set to 0. Finally, the total evaluation metric value is obtained by averaging r_i over all viewpoints. In this way, the proposed metric has the significant advantage of being able to perform evaluations with only single input image.

3 Experimental Results

3.1 Experimental Settings

3.1.1 Dataset Selection. Following previous studies [Wang et al. 2024; Zou et al. 2024], we used 3D models in Google Scanned Objects (GSO) dataset [Downs et al. 2022]. From this dataset, we selected 30 objects, which is the same number as in [Tang et al. 2024, 2023; Wang et al. 2024]. When selecting the data, we set the following three requirements. First, the object must be a single entity. Second, when the object detection is applied, the entire area of the object must be included into the bounding box. Third, the object does not necessarily belong to the classes targeted by the object detection model $\mathcal{M}$ as long as it can be detected as some kind of object. This is because we treat the class information and confidence scores of top five results as features, rather than the object detection results.

3.1.2 Experimental Details. We used the following single image-to-3D models: DreamGaussian (DG) [Tang et al. 2023] (released in Sep. 2023), Triplane Gaussian (TGS) [Zou et al. 2024] (Dec. 2023), LGM [Tang et al. 2024] (Feb. 2024), and CRM [Wang et al. 2024] (Mar. 2024). The resolution of each image was resized to 512×512 pixels. As the object detection model $\mathcal{M}$, we employed YOLOX [Ge et al. 2021], namely the pretrained model YOLOX-x, which has the most parameters and the highest mAP value among the pre-trained YOLOX models. The 3D objects generated by the single image-to-3D models were rendered at 15-degree intervals in both elevation and azimuth, resulting in a total of 266 novel-view images. When calculating our evaluation metric, the temperature parameter of the softmax function was set to 0.3. For comparative metrics, we used PSNR [Wang et al. 2004], CD [Borgefors 1986], Volume IoU (V-IoU), F-score, LPIPS [Zhang et al. 2018] and CLIP-similarity [Radford et al. 2021]. Note that while the proposed metric was calculated

Table 1: Average Spearman’s rank correlation coefficients (ρ) of eight evaluation metrics.

	PSNR	SSIM	CD	V-IoU	F-score	LPIPS	CLIP-sim.	Ours
ρ	0.0467	0.0533	0.187	−0.0600	0.150	0.180	0.307	0.373

using only the input image, all the comparative metrics also used the original 3D models or the reference images, which were additionally rendered from the same angles as the novel-view images. Therefore, the evaluation of the proposed metric was solely conducted under exclusively strict conditions with limited assets. PSNR, SSIM, LPIPS and CLIP-similarity were calculated as the average of the metric values of each novel-view image, in the same manners as the conventional approaches. When calculating CD, V-IoU and F-score, the 3D models were normalized to a range of $[-0.5, 0.5]$. Following [Wang et al. 2024], the threshold for F-score was set to 0.05. Although some single image-to-3D models [Tang et al. 2024, 2023; Zou et al. 2024] can generate 3D models in 3D Gaussian Splatting [Kerbl et al. 2023] representation, all generated models were converted to meshes before evaluations for simplicity.

3.2 Validation of Our Evaluation Metric

First, we conduct the validation of our evaluation metric. We assume that the generation quality tends to correlate with the publication dates of the single image-to-3D models. This order is consistent with the results of the user studies and the evaluations by other metrics conducted in previous studies [Tang et al. 2024; Wang et al. 2024], i.e., the quality increases in order of DreamGaussian [Tang et al. 2023] (Sep. 2023), Triplane Gaussian [Zou et al. 2024] (Dec. 2023), LGM [Tang et al. 2024] (Feb. 2024) and CRM [Wang et al. 2024] (Mar. 2024). Therefore, we calculated the Spearman’s rank correlation coefficient (ρ) [Spearman 1904] between the evaluation results and this order. While we used the reference images or the original 3D models to calculate comparative metrics, our metric is calculated only using one input image. Table 1 shows average Spearman’s rank correlation coefficient for the 30 data. It can be seen that the rank correlation coefficient of the proposed metric is highest among all metrics. This result indicates that the evaluation results of our proposed metric correlates with both the order of the models’ publication dates and the experimental results in the previous studies [Tang et al. 2024; Wang et al. 2024].

Next, we investigate whether our proposed metric can truly account for semantic information. Since CLIP-similarity also considers semantic information and is calculated under ideal conditions using reference images, we calculated the average Pearson’s correlation coefficient [Pearson 1895] between the metric values of our metric and CLIP-similarity. The resulting coefficient is 0.502 ($p < .001$), which confirms that our proposed metric certainly considers semantic information. Although the proposed metric evaluates with just one image while the others also use additional reference materials, our metric achieves more accurate evaluations.

3.3 Examples of Evaluation Results

We also show the examples of evaluation results in Table 2. While most comparative metrics highly evaluate unsuccessful generation results, the proposed metric highly evaluates successful ones. Note that our objective is identify the ranking of each metric value, and

Table 2: Examples of evaluation results. Bold types indicate that each metric has been rated as having better quality.

Input	Successful (TGS [Zou et al. 2024])	Unsuccessful (CRM [Wang et al. 2024])	Input	Successful (CRM [Wang et al. 2024])	Unsuccessful (TGS [Zou et al. 2024])
					
PSNR (↑)	15.313	16.178	PSNR (↑)	15.534	17.551
SSIM (↑)	0.876	0.873	SSIM (↑)	0.866	0.876
CD (↓)	0.00337	0.00296	CD (↓)	0.00506	0.00396
V-IoU (↑)	0.348	0.390	V-IoU (↑)	0.140	0.387
F-score (↑)	0.792	0.855	F-score (↑)	0.612	0.736
LPIPS (↓)	0.178	0.166	LPIPS (↓)	0.181	0.171
CLIP-sim. (↑)	0.926	0.927	CLIP-sim. (↑)	0.884	0.881
Ours (↑)	0.539	0.528	Ours (↑)	0.441	0.320

it is not meaningful to directly compare absolute metric values between different metrics. As for the horse example, the unsuccessful result differs in the shape of the front legs and has an excessive number of hind legs. On the other hand, in the clock example, the dial of the successful result is clear, whereas that of the unsuccessful result is unclear. Our proposed metric assigns higher metric values to the successful generation results, while the comparative metrics often evaluate the unsuccessful ones higher. Therefore, it is confirmed that our proposed metric provides evaluations that align more closely with human perception.

4 Discussion and Future Work

4.1 Discussion

The experimental results have demonstrated that our proposed metric effectively works in evaluating single image-to-3D models. The proposed metric is distinctive as it can perform evaluations focusing on objects. The advantage of the proposed metric is that it can perform evaluations based on a single input image, unlike the conventional metrics that require additional reference images or the original 3D models.

4.2 Future Work

Our proposed metric faces limitations in the input data of single image-to-3D models since it is difficult for object detection models to identify objects that are not presented in their datasets. Also, future work includes the validation of our proposed metric through user study using mean opinion scores and additional experiments on various data containing multiple objects.

5 Conclusion

In this paper, we have proposed a novel evaluation metric for single image-to-3D models based on object detection perspective. The proposed metric is designed to especially focus on objects and consider semantic information, and enables evaluations from only one input image. In addition, we have conducted the validation of our proposed metric from several aspects.

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References

- Gunilla Borgefors. 1986. Distance transformations in digital images. *Computer Vision, Graphics, and Image Processing* 34, 3 (1986), 344–371.
- Kevin Clark, Minh T. Luong, Quoc V. Le, and Christopher D. Manning. 2020. ELECTRA: Pre-training text encoders as discriminators rather than generators. In *Proc. of the International Conference on Learning Representations (ICLR)*.
- Laura Downs, Anthony Francis, Nate Koenig, Brandon Kinman, Ryan Hickman, Krista Reymann, Thomas B. McHugh, and Vincent Vanhoucke. 2022. Google scanned objects: A high-quality dataset of 3D scanned household items. In *Proc. of the International Conference on Robotics and Automation (ICRA)*. 2553–2560.
- Zheng Ge, Songtao Liu, Feng Wang, Zeming Li, and Jian Sun. 2021. YOLOX: Exceeding YOLO series in 2021. *arXiv:2107.08430* (2021).
- Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3D gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics* 42, 4 (2023), 1–14.
- Karl Pearson. 1895. VII. Note on regression and inheritance in the case of two parents. *The Royal Society of London* 58, 347–352 (1895), 240–242.
- Alec Radford, Jong W. Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *Proc. of the International Conference on Machine Learning (ICML)*. 8748–8763.
- Charles Spearman. 1904. The proof and measurement of association between two things. *The American Journal of Psychology* 15, 1 (1904), 72–101.
- Jiaxiang Tang, Zhaoxi Chen, Xiaokang Chen, Tengfei Wang, Gang Zeng, and Ziwei Liu. 2024. LGM: Large multi-view gaussian model for high-resolution 3D content creation. In *Proc. of the European Conference on Computer Vision (ECCV)*. 1–18.
- Jiaxiang Tang, Jiawei Ren, Hang Zhou, Ziwei Liu, and Gang Zeng. 2023. Dreamgaussian: Generative gaussian splatting for efficient 3D content creation. In *Proc. of the International Conference on Learning Representations (ICLR)*.
- Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. 2004. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing* 13, 4 (2004), 600–612.
- Zhengyi Wang, Yikai Wang, Yifei Chen, Chendong Xiang, Shuo Chen, Dajiang Yu, Chongxuan Li, Hang Su, and Jun Zhu. 2024. CRM: Single image to 3D textured mesh with convolutional reconstruction model. In *Proc. of the European Conference on Computer Vision (ECCV)*. 1–19.
- Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. 2018. The unreasonable effectiveness of deep features as a perceptual metric. In *Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 586–595.
- Zi X. Zou, Zhipeng Yu, Yuan C. Guo, Yangguang Li, Ding Liang, Yan P. Cao, and Song H. Zhang. 2024. Triplane meets gaussian splatting: Fast and generalizable single-view 3D reconstruction with transformers. In *Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 10324–10335.